

EDUCATION	University of Pittsburgh Ph.D. Intelligent Systems	2021 - present <i>GPA : 3.7/4.0</i>
	Birla Institute of Technology and Science, Pilani M.E. Software Systems	2018 - 20
	Thesis : Hyperspectral guided Image Dehazing	
SELECTED PUBLICATIONS	B.E. (Hons.) Electrical & Electronics	2013 - 18
	Thesis : Deep Learning for Biometrics	
	RESPLAT: Reversible Anatomy-Consistent 3D Gaussian Editing under Sparse X-ray Supervision <u>H Sinha</u> , A Patil, G Sharma, Kanav, Y Sinha, R Manekar <i>Under Review</i>	
	Soft Hull: 3D Gaussian Splatting Unlearning G Sharma, A Patil, <u>H Sinha</u> , R Manekar, Y Sinha <i>Under Review</i>	
	GS-VO: Causal Metric Planar Odometry with Dual-Gaussian Consensus and Learned Trust A Patil, G Sharma, <u>H Sinha</u> , Y Sinha, R Manekar <i>Under Review</i>	
	Semantically Safe, Interactionally Unsafe: Human Control and Feasibility Limits in Child-Facing AI <u>H Sinha</u> , P Kumar, K Haribabu, V Chamola, D Kumar, A Pal, and Y Sinha <i>Under Review</i>	
	SoK: The Representation–Origin Gap Authentication When Presence Becomes Generatable Neha, Y Sinha, <u>H Sinha</u> , H K, N Chilamkurti <i>Under Review</i>	
	Hybrid novice–AI system achieves expert-level performance in intraoperative ischemia detection N Murali, AI Mina, <u>H Sinha</u> , JW Anderson, Y Raka, H-C Chang, HK Amiri, PD Thirumala, K Batmanghelich, S Visweswaran <i>Under Review</i>	
	spatiAlytica: Viewer-Grounded Multimodal Agentic System for Interactive Spatial Omics Analysis A Das, K Zhang, J Song, M Han, A Chen, W Meng, H Galloway, PY Chen, S Jo, Z Liu, MM Hasib, A Officer, <u>H Sinha</u> , YC Chiu, SJ Gao, L Li, Y Huang <i>Under Review</i>	
	Spatial Single-Cell Atlas Reveals KSHV-Driven Broad Cellular Reprogramming, Progenitor Expansion, Immune and Vascular Remodeling in Kaposi’s Sarcoma W Meng, A Das, <u>H Sinha</u> , R Naous, PM Bracci, M McGrath, Y Huang, SJ Gao <i>Under Review; bioRxiv preprint, 2025</i>	
	Unifying AI-assisted scientific discovery around exploration, hypothesis generation, and testing MM Hasib, S Jo, <u>H Sinha</u> , et al. <i>In Patterns, 101660, 2026 (Review)</i>	

STiLE: Automated Tissue Microarray Dearraying for Spatial Transcriptomics

H Sinha, A Das, YC Chiu, SJ Gao, Y Huang

In Bioinformatics, 2026

Solving the Pervasive Problem of Protocol Non-Compliance in MRI using an Open-Source tool *mrQA*

H Sinha and PR Raamana

In Neuroinformatics, 2024

MRdataset: A unified and user-friendly interface to medical imaging datasets

H Sinha and PR Raamana

In Journal of Open Source Software, 9(95), 6269, 2024

Towards Shape-regularized Learning for Mitigating Texture Bias in CNNs

H Sinha and A Kovashka

In ACM International Conference on Multimedia Retrieval (ICMR 2023) | Oral Session 🎤

Cycle Consistent Embedding of 3D Brains with Auto-Encoding Generative Adversarial Networks

S Xing, H Sinha, and SJ Hwang

In International Conference on Medical Imaging with Deep Learning (MIDL 2021)

Domain-Aware Unsupervised Hyperspectral Reconstruction for Aerial Image Dehazing

A Mehta, H Sinha, M Mandal and P Narang

In IEEE Winter Conference on Applications of Computer Vision (WACV 2021)

HIDEGAN: A Hyperspectral-guided Image Dehazing GAN

A Mehta, H Sinha, P Narang and M Mandal

In IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2020)

Audio Classification using Braided Convolutional Neural Networks

H Sinha, V Awasthi and Pawan K Ajmera

In IET Signal Processing, 2020

Text-Convolutional Neural Networks for Fake News Detection in Tweets.

H Sinha, S Kalra and Y Sharma.

In Evolution in Computational Intelligence, Springer, 2020.

Upgrading Security and Protection in Ear Biometrics

H Sinha and PK Ajmera

In IET Biometrics, 2019

Convolutional Neural Network-based Human Identification using Outer Ear Images

H Sinha, R Manekar, Y Sinha, and Pawan K Ajmera

In Soft Computing for Problem Solving, 2019

Studying the Role of Kinect as a Multi-Sensory Learning Platform for Children

H Sinha, S Srivastava, and Y Sinha

In ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp 2018)

Interweaving Convolutions: An application to Audio Classification

H Sinha and PK Ajmera

In International Conference on Knowledge Discovery and Data Mining, DL Day Workshop (KDD 2018)

ABSTRACTS

A Spatial Single-Cell Atlas of Kaposi's Sarcoma: Insights into KSHV Tropism, Tumor Cell Origins, and Cellular Reprogramming in Tumor Progression, Angiogenesis, and Immune Evasion

W Meng, A Das, [H Sinha](#), M Han, R Naous, PM Bracci, MS McGrath, Y Huang, SJ Gao
In 27TH International Workshop on Kaposi's Sarcoma Herpesvirus (KSHV 2025)

Mapping the spatial biology of Kaposi Sarcoma (KS) in skin: Insights into tumor growth and cell behavior

A Das, W Meng, [H Sinha](#), SJ Gao, Y Huang
In 2024 CSBC / MetNet Annual Investigators Meeting (CSBC 2024)

Indiscriminate Adjustment for Confounders is worse than you think and What can be done about it [H Sinha](#), and PR Raamana

In Alzheimer's Association International Conference (AAIC 2024)

Digging deeper into the pervasive problem of non-compliance in MR datasets

[H Sinha](#), and PR Raamana
In Organization for Human Brain Mapping Annual Meeting (OHBM 2024) | Oral Session 🎤
In ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM 2024)

Solving the Pervasive Problem of Protocol Non-Compliance in MR imaging using an open-source tool mrQA

[H Sinha](#), and PR Raamana
In ISMRM & ISMRT Annual Meeting & Exhibition (ISMRM 2023) | Oral Session 🎤
In Organization for Human Brain Mapping Annual Meeting (OHBM 2023)
In Alzheimer's Association International Conference (AAIC 2023)

Building bridges in MR research: unified interface to different MRI dataset formats with MRdataset

[H Sinha](#), and PR Raamana
In INCF Neuroinformatics Assembly (INCF 2023) | Oral Session 🎤

Bridging the Brain-Age Gap: Quantifying regional contributions using Shapley-Owen values

[H Sinha](#), and PR Raamana
In Alzheimer's Association International Conference (AAIC 2023)

Garbage in, Garbage out : Protocol Compliance, Integrity and Quality Assurance of Neuroimaging Datasets

[H Sinha](#), T Boonchalermvichien and PR Raamana
In INCF Neuroinformatics Assembly (INCF 2022)

Learning to Enhance Visual Quality via Hyperspectral Domain Mapping

[H Sinha](#), A Mehta, M Mandal and P Narang
In AAAI Conference on Artificial Intelligence (AAAI 2021)

TEACHING

University of Pittsburgh

Introduction to Machine Learning	Spring 2024
Introduction to Computing for the Humanities	Fall 2023, Fall 2026
Discrete Structures for CS	Fall 2023, Fall 2026
Introduction to Computing for Scientists	Fall 2026

Johns Hopkins University

Reproducible Research	Summer 2015
Exploratory Data Analysis	Summer 2015

SOFTWARE	<i>mrQA</i> : Automatic protocol compliance checks on MR datasets	 
	<i>MRdataset</i> : Unified interface to neuroimaging datasets	
HONORS & AWARDS	ISMRM & ISMRT Annual Meeting & Exhibition Educational Stipend	2023, 2024
	Intelligent Systems Program Fellowship	2021-2023
	UbiComp 2018 Travel Grant	Oct 2018
	Microsoft Code.Fun.Do. Finalist Forum	Feb 2016 - Aug 2016
REVIEWER	International Conference of IEEE Engineering in Medicine and Biology and Society (EMBC)	2025
	Transactions on Circuits and Systems for Video Technology (IEEE-TCSVT)	2024-25
	Journal of Open Source Software (JOSS)	2024
	International Conference on Biomedical and Health Informatics (IEEE-EMBS BHI)	2024-25
	British Machine Vision Conference (BMVC)	2020
	Winter Conference on Applications of Computer Vision (WACV)	2021-23
	European Conference on Computer Vision (ECCV)	2022
TECHNICAL SKILLS	IET Biometrics	2019-25
	<i>Languages:</i> PyTorch (Geometric, Lightning), R, Keras, Python, SQL, Java, C/C++ <i>Courses:</i> Causal Inference, Machine Learning, Computer Vision, Object Oriented Programming, Multimodal Neuroimaging, Data Mining, Information Retrieval, Artificial Intelligence, Probability and Statistics <i>MOOCs:</i> Data Science Specialization	